Chloe's Biology Final — Cheat Sheet

May 29, 2026 · 1–3 PM · North Gym · 15-min scan only · **SKIP Hardy-Weinberg**

I. SCI. METHOD

- **IV** = what YOU change (X-axis)
- **DV** = what you MEASURE (Y-axis)
- **Control** = no-treatment group
- **Constants** = kept same across all groups

Graphs: bar = categories · line = time/trend · pie = parts of whole.
Hypothesis: If ___, then ___ (testable!)

II. RNA & PROTEIN SYNTHESIS

CENTRAL DOGMA DNA → (transcription, nucleus) → mRNA → (translation, ribosome/cytoplasm) → Protein

Transcription pairs (DNA template → mRNA)

A→**U** · T→A · G→C · C→G (RNA = uracil, not thymine!)

3 differences DNA vs RNA

DNA: double / deoxyribose / T · RNA: single / ribose / U

3 RNAs

mRNA = message · **rRNA** = ribosome · **tRNA** = amino-acid taxi (anticodon)

Codons

AUG = start (Met) · **UAA / UAG / UGA** = stop

Mutations

Type	Effect
Silent sub	Same amino acid → no change
Missense sub	1 amino acid changed
Nonsense sub	Premature STOP → short protein
Insertion/deletion	FRAMESHIFT → all downstream codons wrong

⚠ Frameshift is usually worst — every codon after = gibberish.

III. MEIOSIS

Human: body 2n=46, gametes n=23. Meiosis = 1 diploid → 4 unique haploid.

Phases (memorize order!)

Prophase I (**tetrads form, crossing over!**) → Meta I (independent assortment) → Ana I (homologs split) → Telo I → **NO interphase** → Prophase II → Meta II → Ana II (sisters split) → Telo II = **4 cells**

Mitosis vs Meiosis

	Mitosis	Meiosis
Cells	2 identical	4 unique
Rounds	1	2
Chromosomes	2n	n
Crossing over	No	Yes (Prophase I)

Spermato vs Oogenesis

Sperm: 4 sperm/meiosis · Egg: 1 ovum + 3 polar bodies (all cytoplasm → egg for embryo support)

3 sources of variation

Crossing over · Independent assortment · Random fertilization

IV. GENETICS

Mendel's 3 laws

Dominance: one allele can mask another · **Segregation:** alleles split into separate gametes · **Independent Assortment:** different traits sort independently

Genotype letters

Pattern	What it is
PP	Homozygous dominant
Pp	Heterozygous (hybrid)
pp	Homozygous recessive

Classic ratios

Monohybrid het × het = **3:1** phenotypes (1:2:1 genotypes)
Dihybrid het × het = **9:3:3:1** phenotypes
Incomplete dominance het × het = **1:2:1** (e.g. red:pink:white)

Non-Mendelian

- **Incomplete dom.** = BLENDED (red + white = pink)
- **Codominance** = BOTH shown (AB blood, roan cattle)
- **Multiple alleles** = >2 versions of a gene (blood: I^A, I^B, i)
- **Polygenic** = many genes, bell curve (height)

Blood type genotypes

A = $I^A I^A$ or $I^A i$ · B = $I^B I^B$ or $I^B i$ · AB = $I^A I^B$ · O = ii

X-linked

Recessive X-trait far more common in **males** (only 1 X → no backup). Color-blindness: ~1/12 males, ~1/200 females.

Karyotype

Picture of all chromosomes. XX = ♀, XY = ♂. Detects trisomy (e.g. Down = trisomy 21).

⚠ Incomplete dom = BLENDED; Codom = BOTH AT ONCE.

V. EVOLUTION

Natural selection requires:

- Overproduction (more born than survive)
- Heritable variation
- Variation affects fitness (= reproductive success, NOT strength)

Evidence (memorize 5+)

Biogeography · Fossils · Homologous · Vestigial · Embryology · Molecular (DNA, cytochrome C)

⚠ **Homologous** = same structure, diff function (common ancestor!). **Analogous** = diff structure, same function (independent evolution).

3 types of selection

Directional = curve shifts · **Stabilizing** = curve narrows · **Disruptive** = curve splits

Genetic drift

Bottleneck = population crashes · **Founder** = small group starts new pop

3 reproductive isolations

Behavioral (meadowlark songs) · **Geographic** (canyon) · **Temporal** (different times)

Cladogram

Branch points = common ancestors. Closer = more related.

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VI. Ecology · VII. Human Body

VI. ECOLOGY

Levels of organization
Species → Population → Community → Ecosystem → Biome → Biosphere

Biotic vs Abiotic
Biotic = living · Abiotic = water, sun, soil, temp

Producers vs Consumers
Autotroph = makes own food (plants, photosynthesis) · **Heterotroph** = eats others (animals, fungi)

Consumer types
Herbivore · Carnivore · Omnivore · Scavenger (vulture) · Decomposer (breaks down → detritus) · Detritivore (eats detritus)

10% Rule
Only ~10% of energy passes to next trophic level (rest = heat, metabolism). Limits chains to 4-5 levels.

Energy vs matter
Energy **FLOWS** (one way, lost as heat) · Matter **CYCLES** (carbon, water, nitrogen — biogeochem cycles)

Habitat vs Niche
Habitat = address (WHERE) · Niche = job (ROLE) · **Competitive exclusion:** no 2 species share same niche

3 symbioses

Mutualism	+/+	Bee + flower
Commensalism	+/0	Barnacle on whale
Parasitism	+/−	Tapeworm in dog

⚠ Predation ≠ symbiosis (it's not long-term living together).

Limiting factors
Density-dependent = stronger when crowded (competition, disease, predation) · **Density-indep.** = hits regardless (weather, disasters)

Succession

	Primary	Secondary
Starts	Bare rock, no soil	Soil intact
Pioneers	Lichens, mosses	Grasses, weeds
Speed	Slow (centuries)	Faster (decades)
Trigger	Volcano, glacier	Fire, hurricane, farm

End state = **climax community**

Keystone & Invasive
Keystone = big effect, not abundant (sea otters!) · Invasive = non-native, harmful

Biome
Determined by **temperature + precipitation**

VII. REPRODUCTIVE SYSTEM

Male
Testes (sperm + testosterone, in scrotum = cooler) → vas deferens → prostate adds seminal fluid → urethra → out

Female
Ovaries (eggs + estrogen) → fallopian tubes (**fertilization here**) → uterus (implantation) → cervix → vagina

4 hormones

Hormone	Females	Males
FSH (pit)	Matures follicle	Spermatogenesis
LH (pit)	Triggers ovulation	Triggers testosterone
Estrogen	Oogenesis + uterine lining	—
Progesterone	Thickens uterine lining	—
Testosterone	—	Sperm + 2° sex char.

Menstrual cycle (~28 days)
Follicular → **Ovulation (~day 14)** → Luteal → Menstruation (if no fertilization)

Pregnancy structures
Placenta (mom↔baby transfer) · Umbilical cord (connects to placenta) · Amniotic sac (protects)

VII. CIRCULATORY

4 chambers (key rule)
RIGHT = deoxygenated (to lungs). LEFT = oxygenated (to body).

Chamber	Does what
R atrium	← body (via vena cavae), deox
R ventricle	→ lungs (pulm. artery)
L atrium	← lungs (pulm. veins), ox
L ventricle	→ body (aorta) — strongest!

Blood flow loop
R atrium → tricuspid → R ventricle → pulm. semilunar → pulm. artery → **LUNGS** → pulm. veins → L atrium → bicuspid → L ventricle → aortic SL → aorta → **BODY** → vena cavae → R atrium

2 circuits
Pulmonary = heart↔lungs · **Systemic** = heart↔body

Vessels
Artery = AWAY from heart (thick, pulse) · **Vein** = TO heart (valves) · **Capillary** = exchange (1 cell thick)

⚠ Pulmonary *artery* = DEOXYGENATED! Pulmonary *vein* = oxygenated. Artery/vein = direction (away/to), not oxygen!

Blood parts
Plasma · RBC (hemoglobin → O_2) · WBC (immunity) · Platelets (clot)

VII. RESPIRATORY

Path of O_2 (memorize!)
Nose → pharynx → larynx → trachea → bronchi → bronchioles → **alveoli** (gas exchange!) → capillary → hemoglobin

Why alveoli rock
Huge surface area · 1 cell thick · moist · dense capillaries

Cilia + mucus
Mucus traps junk, cilia sweep it OUT toward throat

Breathing mechanics
Inhale = diaphragm **contracts down** → chest expands → pressure drops → air IN
Exhale = diaphragm relaxes up → pressure rises → air OUT
Controlled by **medulla oblongata**, monitors CO_2

Nose > mouth breathing
Nose warms, moistens, filters air. Mouth skips it.